

SOLUTION PROPOSAL: AI-Powered Prior Authorization Process Optimization

1 Problem & Opportunity

A regional health plan processes 15,000 prior authorization requests per month through manual nurse review. Clinical documentation arrives through fax, PDFs, portals, and uploads—creating a 4.2-day turnaround time and 22% rework rate due to missing information. Leadership wants 40% turnaround reduction without increasing headcount.

Solution: Three AI opportunities implemented as **Claude Skills** (distributed as Plugins for Claude Coworker) to automate routine tasks while maintaining clinical quality and regulatory compliance.

Metric	Current State	Goal (To-Be)
Monthly PA volume	~15,000 requests	~15,000 (no change)
Average turnaround time	4.2 days	2.5 days (40% reduction)
Rework rate	22% (missing info)	<5% (proactive validation)
Auto-adjudication rate	0% (100% manual)	70-80% (routine cases)
Nurse review focus	Data gathering + clinical judgment	Clinical judgment only
Provider status calls	Frequent	Eliminated (automated updates)

2 Three AI Opportunities as Claude Skills

2.1 Opportunity 1: Document Completeness Checker

Problem: 22% rework rate (3,300 cases/month) due to missing information.

Skill Implementation: - Ingests documentation from any channel (fax OCR, PDF, portal, upload) - Extracts structured data (diagnosis, labs, treatment history) - Validates against PA criteria; identifies missing elements - Returns immediate feedback to provider

Plugin Workflow: Provider submits → Skill validates → Complete cases route to review / Incomplete cases return with specific guidance

Impact: Eliminate 22% rework; reduce turnaround by 2-4 days for affected cases; immediate provider feedback

2.2 Opportunity 2: Clinical Guidelines Verification Engine

Problem: Manual guideline verification for 15,000 cases/month is deterministic work that doesn't require human judgment for routine cases.

Skill Implementation: - Loads clinical guidelines for requested treatment - Evaluates patient data against criteria (diagnosis, labs, prior treatments, contraindications) - Applies decision logic (if/then rules, thresholds, temporal requirements) - Produces: APPROVE / DENY (with reason) / ESCALATE (for nurse review) - Generates audit trail for regulatory compliance

Plugin Workflow: Complete request → Skill evaluates → Routine cases auto-adjudicate (70-80%) / Complex cases escalate to nurse with pre-loaded context

Impact: Auto-adjudicate 70-80% of cases; <1 day turnaround for routine approvals; nurses focus on complex clinical judgment

2.3 Opportunity 3: PA Workflow Orchestrator

Problem: Frequent provider status calls consume nurse time; opaque process creates provider frustration.

Skill Implementation: - Tracks PA status through all stages - Sends automated updates at key milestones - Handles routine inquiries; escalates complex ones - Delivers decisions with clinical rationale

Plugin Workflow: Request submitted → Automated confirmation → Status updates at each stage → Decision notification with rationale

Impact: Eliminate routine status calls; improve provider satisfaction; free nurse capacity from administrative tasks

3 Extensibility & Implementation

Extensibility: Modular Skill architecture extends to appeals, utilization review, case management; integrates with EHR/payer platforms; scales via Plugin distribution model.

Implementation: Phase 1 - Validate with historical data | Phase 2 - Pilot in shadow mode | Phase 3 - Measure & scale

Outcome: 40% turnaround reduction, 70-80% auto-adjudication, maintained clinical quality & regulatory compliance—without headcount increase.

Solution Proposal | July 2026